

Pilot Registry Act

(Model statute for a confederation-level jurisdiction with opt-in member polities)

Article 1 — Purpose & Findings

1.1 Purpose. Establish a centralized Registry and governance regime for time-boxed pilots that test LSC/CB/ICB/SB/ISB rails, standards conformance, sanctions-over-shots, replication governance (RGP), search-and-rescue (SAR) compacts, corridor service-level agreements (SLAs), tribunal SLAs, and related implementations.

1.2 Findings.

- a) Pilots require clear eligibility, transparent methods, auditable data, proportionate enforcement, and fixed sunsets.
- b) Multi-implementation standards and reversible remedies reduce systemic risk and enable safe learning.
- c) No collective punishment; sanctions must be specific to pilots, sponsors, or operators.
- d) Exceptional powers must be caged and time-boxed.

Article 2 — Definitions

2.1 “Pilot” means a time-boxed, scoped test of rails, standards, SLAs, or governance mechanisms registered under this Act.

2.2 “Registry” means the centralized record, systems, and officials administering this Act.

2.3 “Sponsor” means an opt-in polity, public agency, or recognized consortium applying for and bearing primary accountability for a Pilot.

2.4 “**Operator**” means any natural or legal person conducting operational activities within a Pilot.

2.5 “**Data Controller**” means the entity determining purposes and means of processing Pilot data.

2.6 “**Ombuds**” means an independent officer empowered to receive complaints, authorize emergency measures under CUCC Cage, and recommend remedies.

2.7 “**Tribunal**” means the bonded, neutral adjudicative body with authority to issue Interim Orders (IO) and Final Orders (FO) under this Act.

2.8 “**Pillar Index (Π)**” means the geometric mean composite of normalized indicators for the Five Pillars, computed per the Methods Annex.

2.9 “**Methods Annex**” means the document required for each Pilot describing indicators, transformations, data sources, Π aggregation, uncertainty bands, sensitivity analysis, and falsifiers.

2.10 “**Falsifier**” means a pre-stated condition or result that, if observed, requires termination, redesign, or downgrade of claims.

2.11 “**Conformance Suite**” means a complete, independently developed implementation and test battery for a critical rail or specification.

2.12 “**Interim Order (IO)**” means a time-bounded tribunal order issued within the IO SLA.

2.13 “**Final Order (FO)**” means a tribunal order resolving a matter on the merits within the FO SLA.

2.14 “**Reversible Tranche**” means a sanction or remedy structured in increments that can be reversed without irreparable harm.

2.15 “**Replication Governance (RGP)**” means controls over physical/digital replication, including lineage, lot control, and teardown.

2.16 “**Lineage**” means signed provenance records linking inputs, versions, and outputs.

2.17 “**Random Teardown**” means the destructive or functional teardown of a random sample of lots or events to verify RGP.

2.18 “**Green Lane**” means a pre-cleared, low-friction path for rescue and safety actions consistent with Rails and AHF.

2.19 “**Emergency in a Cage (CUCC Cage)**” means time-boxed emergency authority with two-key authorization, hard clocks, quorum engines, scope box, public ledgering, and automatic lapse.

Article 3 — Eligibility & Pre-Registration

3.1 Eligibility.

a) Sponsors must be opt-in member polities, recognized intergovernmental bodies, or public-interest consortia with bonding capacity.

b) Operators must be licensed where applicable and named by the Sponsor.

c) Each Pilot shall specify scope, geography, interfaces, and affected populations.

d) Critical control planes must maintain multi-implementation (≥ 3) where applicable.

3.2 Pre-Registration Lead Time. Sponsors shall submit Pre-Registration ≥ 30 days before pilot start.

3.3 Screening. The Registry shall screen for completeness, conflicts with existing orders, and readiness of Methods Annex.

3.4 Pilot Duration. Default ≤ 12 months, renewable once (total ≤ 24 months).

Article 4 — Registration Requirements

4.1 Mandatory Contents. A Registration must include:

a) Sponsor, Operators, Data Controller, contacts.

b) Scope, schedules, exit criteria, and proposed renewal conditions.

c) Methods Annex (Article 5).

- d) RGP plan including **signed lineage trees** and **Random Teardown 1–3%**.
 - e) Conformance plan demonstrating **at least 3 independently developed implementations for any critical rail; grace window 12 months for major spec changes**.
 - f) Corridor SLAs or statement of non-applicability.
 - g) Tribunal SLAs acknowledgment and compliance plan.
 - h) Privacy plan (Article 6).
 - i) Bond posting and insurance if required by regulation.
- 4.2 **Approval & Posting.** Upon approval, the Registry shall post a **Public pilot record ≤ 7 days from approval**, WCAG-AA compliant.
- 4.3 **Reporting Cadence.** **Weekly (ops log), monthly (metrics), quarterly (audit snapshot).**
- 4.4 **Pilot Start.** A Pilot may commence only after posting and bond confirmation.

Article 5 — Methods & Metrics (Π , Uncertainty, Sensitivity, Falsifiers)

- 5.1 **Methods Annex Required.** Each Pilot shall include a Methods Annex that **includes indicator definitions, transformations, data sources, geometric Π aggregation, uncertainty bands, and sensitivity analysis; must include falsifiers**.
- 5.2 **Computation.** Π shall be computed as the geometric mean of normalized Pillar sub-indices on $[0,1]$, with monotone transforms as specified.
- 5.3 **Uncertainty & Sensitivity.** The Annex shall specify confidence bands, bootstrap or equivalent resampling, and one-at-a-time and joint sensitivity to key assumptions.
- 5.4 **Falsifiers.** The Annex shall enumerate falsifiers; observation of any falsifier requires immediate notification to the Registry and Ombuds and triggers review or termination.

5.5 **Comparability.** Indicators and transforms must be stable across reporting periods; changes require redlines and side-by-side restatements.

Article 6 — Data Handling, Privacy & Publication

6.1 **Data Transparency.** Publish de-identified datasets and code within 30 days of each monthly report; privacy thresholds k-anonymity $k \geq 5$ and differential privacy $\epsilon \leq 1$ for person-level data.

6.2 **Access.** Public releases shall be WCAG-AA compliant and deposited in a Registry-designated repository with checksums and schema documentation.

6.3 **Minimization & Purpose Limitation.** Collect only data necessary for stated indicators and governance.

6.4 **Security.** Implement role-based access, encryption at rest and in transit, and tamper-evident logs.

6.5 **Privacy Audits.** Annual, independent privacy audits are mandatory.

6.6 **Deletion Requests.** Where legally applicable, deletion requests honored ≤ 30 days.

Article 7 — Governance & Oversight (Ombuds, Tribunal, Editors)

7.1 **Registry Editors.** Maintain specifications, conformance suites, and posting quality; publish change logs; ensure **multi-implementation** (≥ 3) for critical control planes.

7.2 **Ombuds.** Receive complaints, monitor harms, co-authorize CUCC Cage actions as second key, and recommend remedies.

7.3 **Tribunal Jurisdiction & SLAs.**

a) **Interim Order (IO)** ≤ 72 hours from properly filed petition.

b) **Final Order (FO)** $\leq 14\text{--}30$ days from joinder, with reasoned decision.

c) **Reversible tranche sanctions permitted at IO.**

d) Hearings may be remote; records are public save for sealed privacy segments.

7.4 **Conflicts.** Officials shall disclose conflicts; recusal required where material.

Article 8 — Audits, Inspections & Teardowns

8.1 **Randomized Audits.** $\geq 5\%$ of all pilots per quarter selected by the Registry; results posted.

8.2 **For-Cause Audits.** Initiated on Tribunal order or Ombuds referral.

8.3 **Operational Inspections.** The Registry may conduct announced or unannounced inspections within scope.

8.4 **RGP Lineage & Teardown.**

a) Maintain **signed lineage trees** for covered artifacts and events.

b) Perform **Random Teardown 1–3% of lots or events**; publish findings.

8.5 **Corrective Actions.** Findings trigger a Corrective Action Plan under Article 12.

Article 9 — Incidents, Evidence & Atrocity-Hardened Fallback

9.1 **Incident Disclosure (Golden Chain of Custody).** Initial evidence pack within **72 hours for safety/security incidents.** Evidence packs shall include logs, version hashes, lineage links, and contact trees.

9.2 **AHF & Green Lanes.** Rescue-first Green Lanes must remain open; AHF procedures shall preserve life and halt cascading harm with minimal necessary scope.

9.3 **CUCC Cage for Emergencies.** Any emergency authority exercised under a Pilot must follow CUCC Cage: **two-key authorization (Commander + Ombuds), hard clocks**

(7/30/90 days), quorum engines active (≥ 3 implementations), public ledger, and automatic lapse absent renewal with published reasons.

9.4 **Cooling/Capacity Buffers (if applicable).** Minimum 7 days strategic buffer with rule-based release; AMM + circuit breakers mandated.

9.5 **Corridor SLAs (if applicable).** Capacity elasticity floor $\geq 15\%$ above 95th percentile demand; maintenance via commit-reveal with 14-day reveal window.

Article 10 — Incentives & Preferences (fee rebates, priority access)

10.1 **Fee Rebates.** The Registry may rebate up to a specified percentage of fees for Pilots meeting conformance on first pass and posting complete data on time for three consecutive months.

10.2 **Priority Access.** The Registry may assign priority in shared test corridors, sandboxes, or compute windows to Pilots demonstrating high Π stability and zero material non-conformances.

10.3 **Non-Discrimination.** Incentives are content-neutral and based on transparent criteria posted by the Registry.

Article 11 — Enforcement & Sanctions

11.1 **Principles.** Sanctions are proportionate, reasoned, evidence-based, and specific to the Pilot, Sponsor, or Operator; **no collective punishment.** “Logs before labels” applies—evidence precedes characterization.

11.2 **Sanctions Ladder (mandated).**

a) **Notice of Non-Conformance (public) + Corrective Action Plan ≤ 14 days.**

b) **Administrative Penalty: 1–5% of posted bond; public censure.**

c) **Bond Burn:** additional **5–20%** on repeated or material breach; **pilot pause up to 30 days.**

d) **Suspension/Delisting:** removal from Registry; pilot termination; **sponsor ineligibility 6–24 months.**

e) **Access Sanctions:** denial of port/slot/corridor preferences linked to the pilot; **only by tribunal FO.**

f) **Referral:** fraudulent reporting or willful harm referred for prosecution; civil liability preserved.

(All sanctions proportionate, reasoned, and published with evidence summaries; **appeals permitted to Tribunal under SLAs.**)

11.3 Interim Measures. At IO, the Tribunal may impose **Reversible Tranche** sanctions to prevent irreparable harm.

Article 12 — Funding & Fees

12.1 Funding. The Registry is funded by appropriations, scaled fees, and penalties. Penalties do not fund adjudication functions.

12.2 Fees. The Registry shall publish a fee schedule proportional to Pilot complexity, with waivers for public-interest or SAR pilots.

12.3 Bonds. Sponsors shall post bonds sized to exposure and corridor footprint, callable solely per Article 11.

Article 13 — Sunset & Periodic Review

13.1 Pilot End & Renewal. Each pilot automatically ceases at its end date unless renewal granted (one time).

13.2 Registry Review. Registry undergoes formal review every 24 months.

13.3 Act Sunset. The Pilot Registry Act sunsets 5 years from enactment unless reauthorized after a public methods audit and outcomes review.

13.4 Emergency Powers Sunset. No indefinite emergencies; all exceptional powers must be caged and time-boxed per Article 9.3.

13.5 Reasons. The Registry shall publish reasons for every sanction and renewal.

Article 14 — Accessibility & Publication Standards

14.1 Accessibility. All public outputs under this Act shall meet **WCAG-AA**.

14.2 Versioning. Specifications, conformance suites, and Methods Annex changes shall use semantic versioning with public diffs.

Appendix A — Templates

A.1 Pre-Registration Form (Required Fields).

1. Sponsor; Operator(s); Data Controller.
2. Scope, geography, interface maps.
3. Start date; **Pilot duration** declaration.
4. Methods Annex summary and link.
5. RGP plan (lineage, **Random Teardown 1–3%**).
6. Conformance plan (**≥ 3 implementations; 12-month grace** for major spec changes).
7. Corridor SLA applicability; cooling/capacity buffers.
8. Privacy plan ($k \geq 5$, $\epsilon \leq 1$).
9. Reporting contacts; escalation tree.
10. Bond and insurance details.

A.2 Methods Annex (Structure).

1. Indicator list, units, directions.
2. Normalization transforms to $[0, 1]$.
3. Π computation (geometric mean).

4. Uncertainty bands methodology.
5. Sensitivity analysis plan.
6. Data sources and refresh rates.
7. Falsifiers (with triggers).
8. Auditability checklist.

A.3 Reporting Pack.

- **Weekly Ops Log:** incidents, downtimes, corridor metrics, RGP lots, lineage updates.
- **Monthly Metrics:** indicator table, Π , uncertainty bands, sensitivity deltas; **publish data/code within 30 days.**
- **Quarterly Audit Snapshot:** conformance status, teardown outcomes, CAP status.

A.4 Evidence Pack (Golden Chain of Custody).

- Event timeline; investigators; signed logs; lineage links; version hashes; affected cohorts; remedial actions.
- **Initial evidence pack within 72 hours** for safety/security incidents.

A.5 Corrective Action Plan (CAP).

- Root cause; corrective steps; owners; deadlines; verification tests; re-audit date.

Implementation Checklist (12 steps)

1. **Constitute Registry** (confederation level); appoint Editors, Ombuds, and seat Tribunal panel lists. *(Day 0–30)*
2. **Publish Fee & Bond Schedules** and WCAG-AA template pack (Appendix A). *(Day 30)*
3. **Stand-up Repositories & Ledgers** for public posting, evidence packs, and versioning; enable hashed lineage store. *(Day 30–45)*
4. **Adopt Conformance Suites** for first critical rails with **≥ 3 independently developed implementations**; publish 12-month grace policy. *(Day 45–60)*
5. **Issue Pre-Registration Call** with **≥ 30 days lead time**; open screening queue. *(Day 60)*

6. **Train Case Officers** on IO/FO **SLAs ($\leq 72\text{h}$ / $\leq 14\text{--}30\text{d}$)**, reversible tranches, and AHF/CUCC Cage procedures. *(Day 60–75)*
7. **Finalize Audit Program** to sample **$\geq 5\%$ pilots per quarter**; tool up for for-cause audits. *(Day 75)*
8. **Deploy RGP Controls**: lineage signing, **Random Teardown 1–3%**, and teardown labs. *(Day 75–90)*
9. **Publish Privacy Controls**: implement $k \geq 5$ and $\epsilon \leq 1$ defaults; schedule **annual independent privacy audits** and **$\leq 30\text{-day}$ deletion** workflow. *(Day 90)*
10. **Activate Reporting Cadence**: weekly ops logs, monthly metrics with **data/code publication ≤ 30 days**, quarterly audit snapshots. *(Pilot start + ongoing)*
11. **Stand-up Incident Response**: 24/7 intake; **initial evidence pack ≤ 72 hours**; Golden Chain of Custody procedures. *(Go-live)*
12. **Schedule Reviews**: **Registry review every 24 months**; docket Act sunset briefing before Year 5; publish reasons for all sanctions and renewals. *(Ongoing)*

Optional Knobs (policy switches before launch)

- **Jurisdiction Variant**: apply as a “city-state,” “federal ministry,” or “intergovernmental compact.”
- **Audit Rate**: increase randomized audits from **5% $\rightarrow$ 10%** during early rollout.
- **Privacy Floor**: tighten to **$k \geq 10$** and **$\epsilon \leq 0.5$** for highly sensitive data.
- **Pilot Cap**: set “no more than N concurrent pilots per sponsor.”

End of Model Statute.